

## Algorithm for the False-Position Method (Regula Falsi)

### Step 1: Input Parameters

- Choose two initial guesses:  $x_0$  and  $x_1$ .
- Set the maximum number of iterations.
- Set the error tolerance.

### Step 2: Check if a Root Exists

- If  $f(x_0) \times f(x_1) \geq 0$ , there is no guarantee of a root in the given interval. Display a warning and stop execution.

### Step 3: Initialize Variables

- Set iteration counter  $n=1$ .
- Initialize error  $er=1$

### Step 4: Iteration Process

- While  $n \leq \text{step}$  and  $er > \text{epes}$ :
  1. Compute the **False-Position formula**:

$$x_2 = x_1 - \frac{f(x_1) * (x_1 - x_0)}{f(x_1) - f(x_0)}$$

2. Compute the relative error:  $er = |x_2 - x_1|/x_2$  this is correct ->  $er = |(x_2 - x_{\text{old}})/x_2$
3. Update the interval:
  - If  $f(x_0) \times f(x_2) < 0$ , set  $x_1 = x_2$ .
  - Else, set  $x_0 = x_2$ .
4. Increment  $n$ .

### Example Walkthrough

Given  $f(x)=x^2-4$ , with initial guesses  $x_0=1$  and  $x_1=3$ :

$$f(1) = -3 \quad f(3) = 5$$

$$f(1) * f(3) < 0$$

find  $x_2$

$$x_2 = x_1 - \frac{f(x_1) * (x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = 3 - \frac{5 * (3 - 1)}{5 - (-3)}$$

$$x_2 = 3 - \frac{10}{8}$$

$$x_2 = 3 - 1.25$$

$$x_2 = 1.75$$

$$\begin{aligned} \text{find, } f(x_2) &= 1.75^2 - 4 \\ &= -0.9375 \end{aligned}$$

Then, check

$$F(x_0) * f(x_2) < 0$$

$$(-) * (-) > 0$$

So,

Else

$$x_0 = x_2$$

$$X_0 = 1.75$$

Again, find  $x_2$